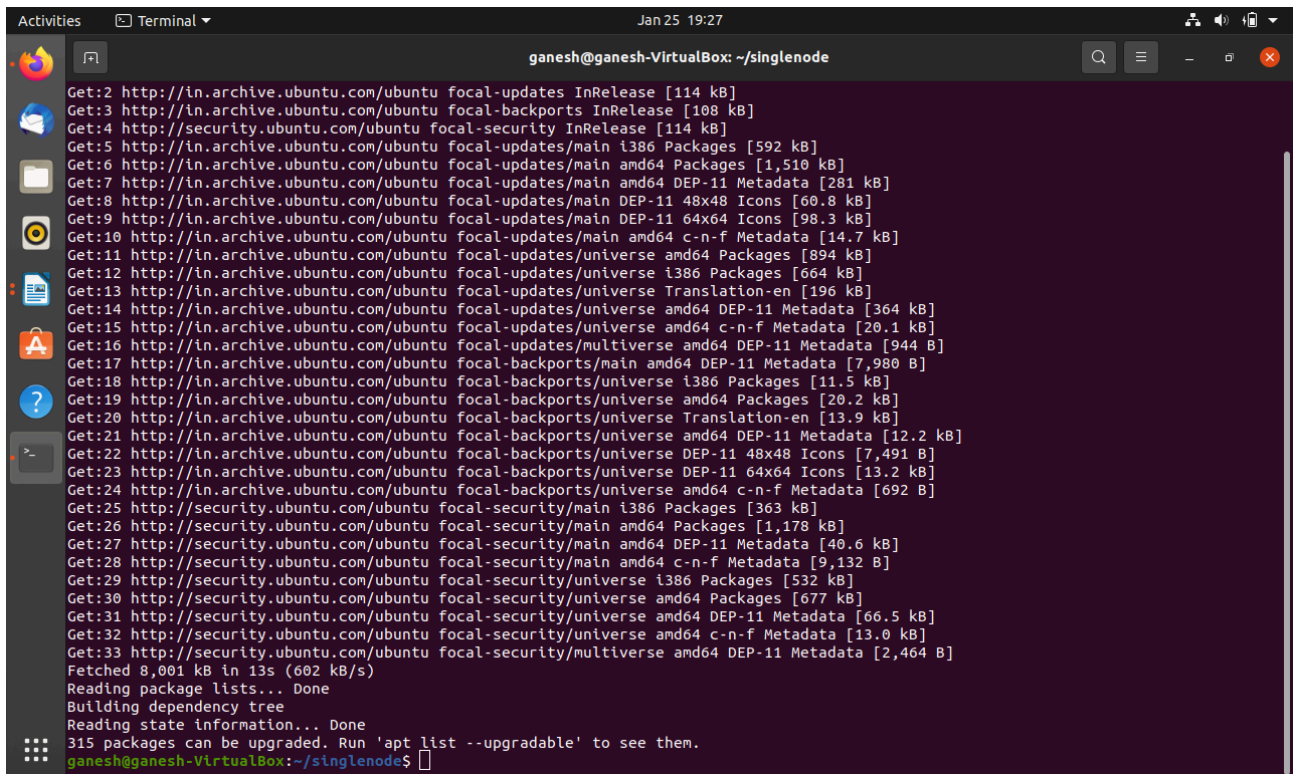


1. Update the repositories
sudo apt update

A screenshot of a terminal window titled 'ganesh@ganesh-VirtualBox: ~/singlenode'. The terminal shows the output of the 'sudo apt update' command. It lists 33 packages being updated from various sources like 'in.archive.ubuntu.com' and 'security.ubuntu.com'. The output includes package names, versions, and sizes. At the bottom, it states '315 packages can be upgraded. Run 'apt list --upgradable' to see them.'

```
Get:2 http://in.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu focal-backports InRelease [108 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [592 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [1,510 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 DEP-11 Metadata [281 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu focal-updates/main DEP-11 48x48 Icons [60.8 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu focal-updates/main DEP-11 64x64 Icons [98.3 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [14.7 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [894 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [664 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [196 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 DEP-11 Metadata [364 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [20.1 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 DEP-11 Metadata [944 B]
Get:17 http://in.archive.ubuntu.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [7,980 B]
Get:18 http://in.archive.ubuntu.com/ubuntu focal-backports/universe i386 Packages [11.5 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [20.2 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [13.9 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [12.2 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 48x48 Icons [7,491 B]
Get:23 http://in.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 64x64 Icons [13.2 kB]
Get:24 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [692 B]
Get:25 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [363 kB]
Get:26 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [1,178 kB]
Get:27 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [40.6 kB]
Get:28 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [9,132 B]
Get:29 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [532 kB]
Get:30 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [677 kB]
Get:31 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [66.5 kB]
Get:32 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [13.0 kB]
Get:33 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [2,464 B]
Fetched 8,001 kB in 13s (602 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
315 packages can be upgraded. Run 'apt list --upgradable' to see them.
ganesh@ganesh-VirtualBox:~/singlenode$
```

2. Download the geth 1.10.8 stable file(.tar.gz)
3. Example: extract it using
sudo tar -xvf geth-alltools-linux-amd64-1.10.8-26675454.tar.gz
4. Step into extracted folder
cd geth-alltools-linux-amd64-1.10.8-26675454
5. It should contain the geth file, make it executable with the below command.
sudo chmod +x geth
6. Copy file to the user bin.
sudo cp geth /usr/local/bin/
cd ..
7. Check the Geth version with the following command
geth version
->Version: 1.8.16-stable

```
Activities Terminal Jan 25 19:29
ganesh@ganesh-VirtualBox: ~/singlenode

Get:12 http://in.archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [664 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [196 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 DEP-11 Metadata [364 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [20.1 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 DEP-11 Metadata [944 B]
Get:17 http://in.archive.ubuntu.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [7,980 B]
Get:18 http://in.archive.ubuntu.com/ubuntu focal-backports/universe i386 Packages [11.5 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [20.2 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [13.9 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [12.2 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 48x48 Icons [7,491 B]
Get:23 http://in.archive.ubuntu.com/ubuntu focal-backports/universe DEP-11 64x64 Icons [13.2 kB]
Get:24 http://in.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [692 B]
Get:25 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [363 kB]
Get:26 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [1,178 kB]
Get:27 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [40.6 kB]
Get:28 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [9,132 B]
Get:29 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [532 kB]
Get:30 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [677 kB]
Get:31 http://security.ubuntu.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [66.5 kB]
Get:32 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [13.0 kB]
Get:33 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [2,464 B]
Fetched 8,001 kB in 13s (602 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
315 packages can be upgraded. Run 'apt list --upgradable' to see them.
ganesh@ganesh-VirtualBox:~/singlenode$ geth version
Geth
Version: 1.10.8-stable
Git Commit: 26675454bf93bf904be7a43cce6b3f550115ff90
Git Commit Date: 20210824
Architecture: amd64
Go Version: go1.16.4
Operating System: linux
GOPATH=
GOROOT=go
ganesh@ganesh-VirtualBox:~/singlenode$
```

8. Make a directory
mkdir singlenode

9. go to that directory
cd singlenode

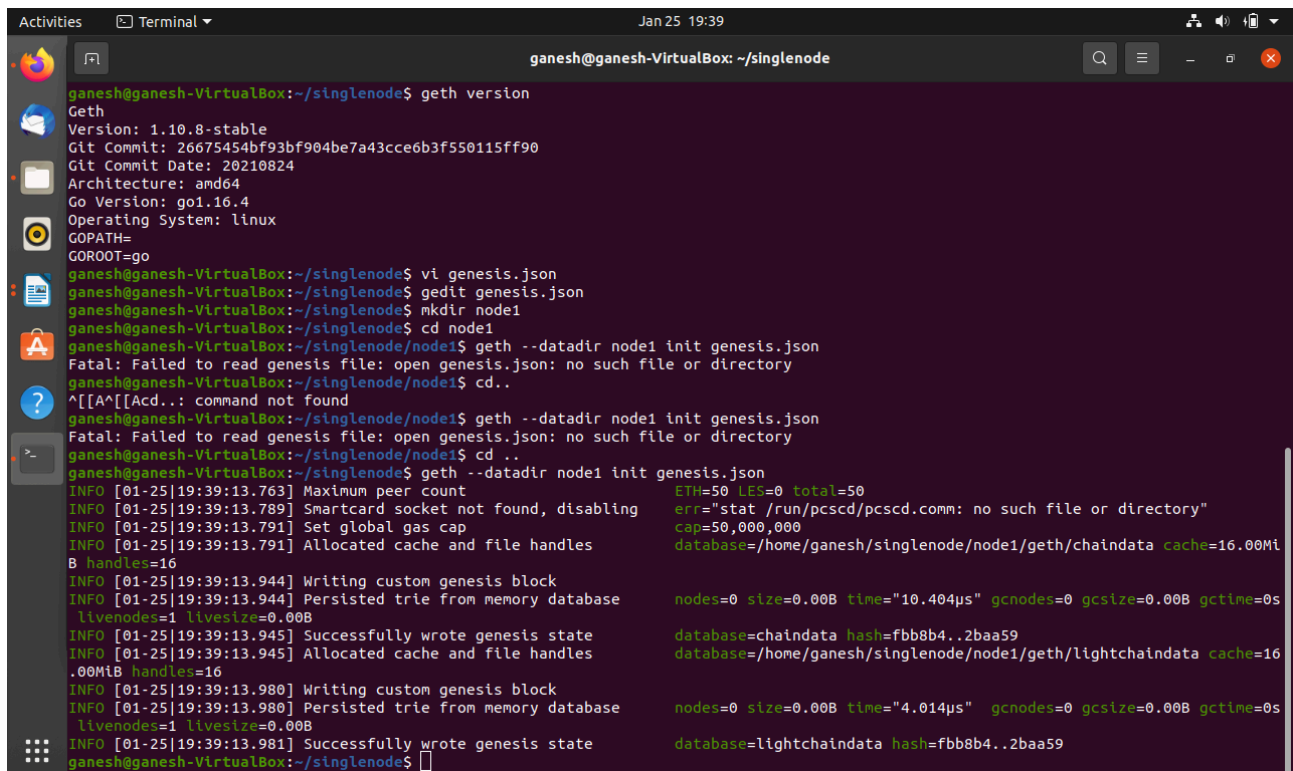
10. create a genesis file (gedit genesis.json)

```
{
"config": {
"chainId": 4321,
"homesteadBlock": 0,
"eip150Block": 0,
"eip155Block": 0,
"eip158Block": 0,
"byzantiumBlock": 0,
"constantinopleBlock": 0
}, "alloc": {},
"difficulty" : "0x20000",
"gasLimit" : "0x8880000"
}
```

```
Activities Text Editor Jan 25 19:31
genesis.json ~/singlenode

Get:1 1
Get:1 2 "config": {
Get:1 3 "chainId": 4321,
Get:1 4 "homesteadBlock": 0,
Get:1 5 "eip150Block": 0,
Get:1 6 "eip155Block": 0,
Get:2 7 "eip158Block": 0,
Get:2 8 "byzantiumBlock": 0,
Get:2 9 "constantinopleBlock": 0
Get:2 10 }, "alloc": {},
Get:2 11 "difficulty" : "0x20000",
Get:2 12 "gasLimit" : "0x8880000"
Get:2 13
Get:2
Get:2
Get:2
Get:3
Get:3
```

11. Create a directory for a node(node 1)
 mkdir node1
12. Initialize genesis to that node using
 geth --datadir node1 init genesis.json



```
ganesh@ganesh-VirtualBox: ~/singlenode
ganesh@ganesh-VirtualBox:~/singlenode$ geth version
Geth
Version: 1.10.8-stable
Git Commit: 26675454bf93bf904be7a43cce6b3f550115ff90
Git Commit Date: 20210824
Architecture: amd64
Go Version: go1.16.4
Operating System: linux
GOPATH=
GOROOT=go
ganesh@ganesh-VirtualBox:~/singlenode$ vi genesis.json
ganesh@ganesh-VirtualBox:~/singlenode$ gedit genesis.json
ganesh@ganesh-VirtualBox:~/singlenode$ mkdir node1
ganesh@ganesh-VirtualBox:~/singlenode$ cd node1
ganesh@ganesh-VirtualBox:~/singlenode/node1$ geth --datadir node1 init genesis.json
Fatal: Failed to read genesis file: open genesis.json: no such file or directory
ganesh@ganesh-VirtualBox:~/singlenode/node1$ cd..
ganesh@ganesh-VirtualBox:~/singlenode$ geth --datadir node1 init genesis.json
Fatal: Failed to read genesis file: open genesis.json: no such file or directory
ganesh@ganesh-VirtualBox:~/singlenode/node1$ cd ..
ganesh@ganesh-VirtualBox:~/singlenode$ geth --datadir node1 init genesis.json
INFO [01-25|19:39:13.763] Maximum peer count                       ETH=50 LES=0 total=50
INFO [01-25|19:39:13.789] Smartcard socket not found, disabling   err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [01-25|19:39:13.791] Set global gas cap                      cap=50,000,000
INFO [01-25|19:39:13.791] Allocated cache and file handles       database=/home/ganesh/singlenode/node1/geth/chaindata cache=16.00MiB handles=16
INFO [01-25|19:39:13.944] Writing custom genesis block
INFO [01-25|19:39:13.944] Persisted trie from memory database    nodes=0 size=0.00B time="10.404µs" gcnodes=0 gcsz=0.00B gctime=0s livenodes=1 liveness=0.00B
INFO [01-25|19:39:13.945] Successfully wrote genesis state       database=chaindata hash=fbb8b4..2baa59
INFO [01-25|19:39:13.945] Allocated cache and file handles       database=/home/ganesh/singlenode/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [01-25|19:39:13.980] Writing custom genesis block
INFO [01-25|19:39:13.980] Persisted trie from memory database    nodes=0 size=0.00B time="4.014µs" gcnodes=0 gcsz=0.00B gctime=0s livenodes=1 liveness=0.00B
INFO [01-25|19:39:13.981] Successfully wrote genesis state       database=lightchaindata hash=fbb8b4..2baa59
ganesh@ganesh-VirtualBox:~/singlenode$
```

13. Now start the node using
 geth --http --http.corsdomain http://remix.ethereum.org --allow-insecure-unlock
 --http --http.port 8545 --http.addr 127.0.0.1 --http.corsdomain "" --http.api
 "eth,net,web3,personal,miner" --datadir node1 --nodiscover --networkid 4321 --port
 30303 console

```
Activities Terminal Jan 25 19:40
ganesh@ganesh-VirtualBox: ~/singlenode

INFO [01-25]19:40:24.548] Allocated cache and file handles database=/home/ganesh/singlenode/node1/ethash/chaindata cache=512.00M
iB handles=524,288
INFO [01-25]19:40:24.634] Opened ancient database database=/home/ganesh/singlenode/node1/ethash/chaindata/ancient reado
nly=false
INFO [01-25]19:40:24.636] Initialised chain configuration config="[ChainID: 4321 Homestead: 0 DAO: <nil> DAOsupport: false EI
P150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: 0 Petersburg: <nil> Istanbul: <nil>, Muir Glacier: <nil>, Berlin: <nil>, Lond
on: <nil>, Engine: unknown]"
INFO [01-25]19:40:24.636] Disk storage enabled for ethash caches dir=/home/ganesh/singlenode/node1/ethash count=3
INFO [01-25]19:40:24.636] Disk storage enabled for ethash DAGs dir=/home/ganesh/.ethash count=2
INFO [01-25]19:40:24.637] Initialising Ethereum protocol network=4321 dbversion=<nil>
INFO [01-25]19:40:24.639] Loaded most recent local header number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
INFO [01-25]19:40:24.639] Loaded most recent local full block number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
INFO [01-25]19:40:24.639] Loaded most recent local fast block number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
WARN [01-25]19:40:24.639] Failed to load snapshot, regenerating err="missing or corrupted snapshot"
INFO [01-25]19:40:24.639] Rebuilding state snapshot
INFO [01-25]19:40:24.642] Regenerated local transaction journal transactions=0 accounts=0
INFO [01-25]19:40:24.648] Resuming state snapshot generation root=56e81f..63b421 accounts=0 slots=0 storage=0.00B elapsed=8.285m
s
INFO [01-25]19:40:24.654] Generated state snapshot accounts=0 slots=0 storage=0.00B elapsed=14.698ms
INFO [01-25]19:40:24.642] Gasprice oracle is ignoring threshold set threshold=2
WARN [01-25]19:40:24.655] Error reading unclean shutdown markers error="leveldb: not found"
INFO [01-25]19:40:24.655] Starting peer-to-peer node instance=Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
INFO [01-25]19:40:24.716] New local node record seq=1 id=7073a317f9e0b7a5 lp=127.0.0.1 udp=0 tcp=30303
INFO [01-25]19:40:24.732] Started P2P networking self="enode://b2bb236206f80170843334e164be2f3785156218c2483744f911e
caca6210cf6a5002c1eddef5e3301659e79821d15d41b27790a74e3a67ff3567e61496a037b@127.0.0.1:30303?discport=0"
INFO [01-25]19:40:24.725] IPC endpoint opened url=/home/ganesh/singlenode/node1/ethash.ipc
INFO [01-25]19:40:24.734] HTTP server started endpoint=127.0.0.1:8545 prefix= cors=http://remix.ethereum.org vhos
ts=localhost reqid=3 t="31.198µs" err="etherbase must be explicitly specified"
WARN [01-25]19:40:24.993] Served eth_coinbase
Welcome to the Geth JavaScript console!

instance: Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
at block: 0 (Thu Jan 01 1970 05:30:00 GMT+0530 (IST))
datadir: /home/ganesh/singlenode/node1
modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-c
>
```

14. now create first account in the node

personal.newAccount()

14*. now create second account in the node

personal.newAccount()

15. set a password

eth.accounts

["0x6e3f9fff3bd1a221f949d6ff320f2fe720daded5",
"0x1b7d8c6763bcd39aa3bb8dfedd0fa992d6a3b2e2"]

```
Activities Terminal Jan 25 19:42
ganesh@ganesh-VirtualBox: ~/singlenode

nly=false
INFO [01-25]19:40:24.636] Initialised chain configuration config="{ChainID: 4321 Homestead: 0 DAO: <nil> DAOSupport: false EI
P150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: 0 Petersburg: <nil> Istanbul: <nil>, Muir Glacier: <nil>, Berlin: <nil>, Lond
on: <nil>, Engine: unknown}"
INFO [01-25]19:40:24.636] Disk storage enabled for ethash caches dir=/home/ganesh/singlenode/node1/ethash count=3
INFO [01-25]19:40:24.636] Disk storage enabled for ethash DAGs dir=/home/ganesh/.ethash count=2
INFO [01-25]19:40:24.637] Initialising Ethereum protocol network=4321 dbversion=<nil>
INFO [01-25]19:40:24.639] Loaded most recent local header number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
INFO [01-25]19:40:24.639] Loaded most recent local full block number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
INFO [01-25]19:40:24.639] Loaded most recent local fast block number=0 hash=fbb8b4..2baa59 td=131,072 age=52y9mo3w
WARN [01-25]19:40:24.639] Failed to load snapshot, regenerating err="missing or corrupted snapshot"
INFO [01-25]19:40:24.639] Rebuilding state snapshot transactions=0 accounts=0
INFO [01-25]19:40:24.642] Regenerated local transaction journal root=56e81f..63b421 accounts=0 slots=0 storage=0.00B elapsed=8.285m
INFO [01-25]19:40:24.648] Resuming state snapshot generation accounts=0 slots=0 storage=0.00B elapsed=14.698ms
INFO [01-25]19:40:24.654] Generated state snapshot threshold=2
INFO [01-25]19:40:24.642] Gasprice oracle is ignoring threshold set error="leveldb: not found"
WARN [01-25]19:40:24.655] Error reading unclean shutdown markers instance=Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
INFO [01-25]19:40:24.655] Starting peer-to-peer node seq=1 ld=7073a317f9e0b7a5 lp=127.0.0.1 udp=0 tcp=30303
INFO [01-25]19:40:24.716] New local node record self="enode://b2bb236206f80170843334e164be2f3785156218c2483744f911e
INFO [01-25]19:40:24.732] Started P2P networking caca6210cf6a5002c1eddef5e3301659e79821d15d41b27790a74e3a67ff3567e61496a037b@127.0.0.1:30303?discport=0"
INFO [01-25]19:40:24.725] IPC endpoint opened url=/home/ganesh/singlenode/node1/ethash.ipc
INFO [01-25]19:40:24.734] HTTP server started endpoint=127.0.0.1:8545 prefix= cors=http://remix.ethereum.org vhos
ts=localhost reqid=3 t="31.198µs" err="etherbase must be explicitly specified"
WARN [01-25]19:40:24.993] Served eth_coinbase
Welcome to the Geth JavaScript console!

instance: Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
at block: 0 (Thu Jan 01 1970 05:30:00 GMT+0530 (IST))
datadir: /home/ganesh/singlenode/node1
modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d
> personal.newAccount()
Passphrase:
Repeat passphrase:
█
```

16. set the miner to the first account

```
miner.setEtherbase(eth.accounts[0])
```

17. start the miner. Here miner started with 1 thread. With 1 thread it is slow. We can also start with

4 threads.

```
miner.start(1)
```

```
Activities Terminal Jan 25 19:49
ganesh@ganesh-VirtualBox: ~/singlenode

INFO [01-25]19:40:24.654] Generated state snapshot accounts=0 slots=0 storage=0.00B elapsed=14.698ms
INFO [01-25]19:40:24.642] Gasprice oracle is ignoring threshold set threshold=2
WARN [01-25]19:40:24.655] Error reading unclean shutdown markers error="leveldb: not found"
INFO [01-25]19:40:24.655] Starting peer-to-peer node instance=Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
INFO [01-25]19:40:24.716] New local node record seq=1 id=7073a317f9e0b7a5 ip=127.0.0.1 udp=0 tcp=30303
INFO [01-25]19:40:24.732] Started P2P networking self="enode://b2bb236206f80170843334e164be2f3785156218c2483744f911e
caca6210cf6a5002c1eddef5e3301659e79821d15d41b27790a74e3a67ff3567e61496a037b@127.0.0.1:30303?discport=0"
INFO [01-25]19:40:24.725] IPC endpoint opened url=/home/ganesh/singlenode/node1/geth.ipc
INFO [01-25]19:40:24.734] HTTP server started endpoint=127.0.0.1:8545 prefix= cors=http://remix.ethereum.org vhos
ts=localhost reqid=3 t="31.198µs" err="etherbase must be explicitly specified"
WARN [01-25]19:40:24.993] Served eth_coinbase
Welcome to the Geth JavaScript console!

instance: Geth/v1.10.8-stable-26675454/linux-amd64/go1.16.4
at block: 0 (Thu Jan 01 1970 05:30:00 GMT+0530 (IST))
datadir: /home/ganesh/singlenode/node1
modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d
> personal.newAccount()
Passphrase:
Repeat passphrase:
INFO [01-25]19:42:30.706] Your new key was generated address=0x9Cdaeba50e1e7Df1690B80385Dd5fEA8d7c8be22
WARN [01-25]19:42:30.713] Please backup your key file! path=/home/ganesh/singlenode/node1/keystore/UTC--2022-01-25T14-12-1
6.963599947Z--9Cdaeba50e1e7Df1690B80385Dd5fEA8d7c8be22
WARN [01-25]19:42:30.713] Please remember your password!
0x9Cdaeba50e1e7Df1690B80385Dd5fEA8d7c8be22
> miner.setEtherbase(eth.accounts[0])
true
> eth.getBalance("0x9Cdaeba50e1e7Df1690B80385Dd5fEA8d7c8be22")
0
> miner.start()
INFO [01-25]19:49:15.814] Updated mining threads threads=1
INFO [01-25]19:49:15.814] Transaction pool price threshold updated price=1,000,000,000
null
> INFO [01-25]19:49:15.820] Commit new mining work number=1 sealhash=22f93b..02ed94 uncles=0 txs=0 gas=0 fees=0 elap
sed=4.685ms
```

18. stop the miner after mining potential block \\\wait for some time before stopping miner
miner.stop()
19. check the balance using
eth.getBalance(eth.accounts[0])
20. set the miner to the second account
miner.setEtherbase(eth.accounts[1])
21. start the miner. Here miner started with 1 thread. With 1 thread it is slow. We can also start with
4 threads.
miner.start(1)
22. stop the miner after mining potential block \\\wait for some time before stopping miner
miner.stop()
23. check the balance using
eth.getBalance(eth.accounts[1])


```

ganesh@ganesh-VirtualBox: ~/singlenode
INFO [01-25|20:54:50.414] Generating DAG in progress epoch=0 percentage=81 elapsed=19m8.421s
INFO [01-25|20:55:11.992] Generating DAG in progress epoch=0 percentage=82 elapsed=19m29.999s
INFO [01-25|20:55:32.900] Generating DAG in progress epoch=0 percentage=83 elapsed=19m50.907s
INFO [01-25|20:55:46.715] Generating DAG in progress epoch=0 percentage=84 elapsed=20m4.722s
INFO [01-25|20:55:58.561] Generating DAG in progress epoch=0 percentage=85 elapsed=20m16.568s
INFO [01-25|20:56:10.281] Generating DAG in progress epoch=0 percentage=86 elapsed=20m28.288s
INFO [01-25|20:56:22.635] Generating DAG in progress epoch=0 percentage=87 elapsed=20m40.642s
INFO [01-25|20:56:34.453] Generating DAG in progress epoch=0 percentage=88 elapsed=20m52.460s
INFO [01-25|20:56:46.473] Generating DAG in progress epoch=0 percentage=89 elapsed=21m4.480s
INFO [01-25|20:56:58.033] Generating DAG in progress epoch=0 percentage=90 elapsed=21m16.040s
INFO [01-25|20:57:09.686] Generating DAG in progress epoch=0 percentage=91 elapsed=21m27.693s
INFO [01-25|20:57:23.623] Generating DAG in progress epoch=0 percentage=92 elapsed=21m41.630s
INFO [01-25|20:57:43.196] Generating DAG in progress epoch=0 percentage=93 elapsed=22m1.203s
INFO [01-25|20:58:01.123] Generating DAG in progress epoch=0 percentage=94 elapsed=22m19.130s
INFO [01-25|20:58:19.244] Generating DAG in progress epoch=0 percentage=95 elapsed=22m37.251s
INFO [01-25|20:58:34.233] Generating DAG in progress epoch=0 percentage=96 elapsed=22m52.240s
INFO [01-25|20:58:47.147] Generating DAG in progress epoch=0 percentage=97 elapsed=23m5.154s
INFO [01-25|20:58:58.981] Generating DAG in progress epoch=0 percentage=98 elapsed=23m16.988s
INFO [01-25|20:59:11.373] Generating DAG in progress epoch=0 percentage=99 elapsed=23m29.380s
INFO [01-25|20:59:11.378] Generated ethash verification cache epoch=0 elapsed=23m29.385s
INFO [01-25|20:59:14.648] Generating ethash verification cache epoch=1 percentage=72 elapsed=3.046s
INFO [01-25|20:59:15.757] Generated ethash verification cache epoch=1 elapsed=4.154s
INFO [01-25|20:59:34.222] Successfully sealed new block number=1 sealhash=dcaa35..a3d272 hash=c17f0f..a6f40e elapsed=23m54.486s
INFO [01-25|20:59:34.222] mined potential block number=1 hash=c17f0f..a6f40e
INFO [01-25|20:59:34.226] Commit new mining work number=2 sealhash=d60297..86e5d7 uncles=0 txs=0 gas=0 fees=0 elapse
d="203.745µs"
INFO [01-25|20:59:41.114] Generating DAG in progress epoch=1 percentage=0 elapsed=25.357s
> miner.stop()INFO [01-25|20:59:48.903] Successfully sealed new block number=2 sealhash=d60297..86e5d7 hash=80777c..f77cf7
elapsed=14.677s
INFO [01-25|20:59:48.903] mined potential block number=2 hash=80777c..f77cf7
INFO [01-25|20:59:48.903] Commit new mining work number=3 sealhash=066daa..dfb19f uncles=0 txs=0 gas=0 fees=0 elapse
d="308.554µs"

null
> eth.getBalance("0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22")
40000000000000000000
>

```

24. Unlock the account from which you want to send funds, to perform the transaction from that account. Here in below last parameter 600 specifies the in seconds for unlock duration.

`web3.personal.unlockAccount(web3.personal.listAccounts[0], null, 6000)`

```

ganesh@ganesh-VirtualBox: ~/singlenode
> web3.eth.sendTransaction({from: "0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22", gasPrice: "2000000000", gas: "2100
0", to: "0x62b80603cc312da015d65194ceb6f5dbd73e0696", value: "1000000000000000000", data: "" })
WARN [01-30|14:16:43.026] Served eth_sendTransaction reqId=12 t="128.519µs" err="authentication neede
d: password or unlock"
Error: authentication needed: password or unlock
    at web3.js:6357:37(47)
    at web3.js:5091:62(37)
    at <eval>:1:25(17)

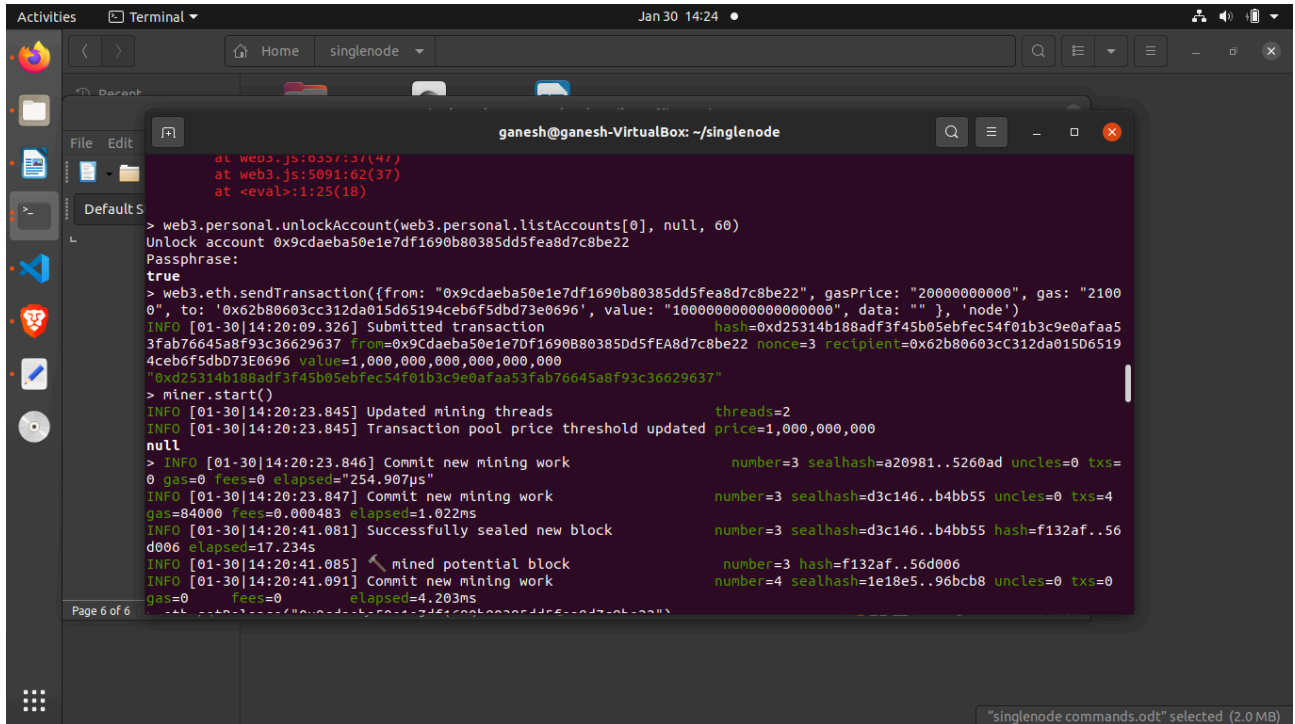
> web3.eth.sendTransaction({from: "0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22", gasPrice: "2000000000", gas: "2100
0", to: "0x62b80603cc312da015d65194ceb6f5dbd73e0696", value: "1000000000000000000", data: "" }, 'node')
WARN [01-30|14:17:29.588] Served eth_sendTransaction reqId=13 t="64.549µs" err="authentication neede
d: password or unlock"
Error: authentication needed: password or unlock
    at web3.js:6357:37(47)
    at web3.js:5091:62(37)
    at <eval>:1:25(18)

> web3.personal.unlockAccount(web3.personal.listAccounts[0], null, 60)
Unlock account 0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22
Passphrase:
true
> web3.eth.sendTransaction({from: "0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22", gasPrice: "2000000000", gas: "2100
0", to: "0x62b80603cc312da015d65194ceb6f5dbd73e0696", value: "1000000000000000000", data: "" }, 'node')
INFO [01-30|14:20:09.326] Submitted transaction hash=0xd25314b188adf3f45b05ebfec54f01b3c9e0afaa5
3fab76645a8f93c36629637 from=0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22 nonce=3 recipient=0x62b80603cc312da015d6519
4ceb6f5dbd73e0696 value=1,000,000,000,000,000,000

```

25. Now send the funds using

```
eth.sendTransaction({from: "0xcd877363343ae452b6de80b460711fdc56ec2eb7",to:
"0x2907068a1836394d1ae345124dbb10dccbf3a0d7", value: "74000000"})
or
eth.sendTransaction({from: eth.accounts[0],to: eth.accounts[1],value: "74000000"})
```



```
ganesh@ganesh-VirtualBox: ~/singlenode
> web3.js:6357:37(47)
at web3.js:5091:62(37)
at <eval>:1:25(18)
> web3.personal.unlockAccount(web3.personal.listAccounts[0], null, 60)
Unlock account 0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22
Passphrase:
true
> web3.eth.sendTransaction({from: "0x9cdaeba50e1e7df1690b80385dd5fea8d7c8be22", gasPrice: "20000000000", gas: "2100
0", to: "0x62b80603cc312da015d65194ceb6f5dbd73e0696", value: "1000000000000000000", data: "" }, 'node')
INFO [01-30|14:20:09.326] Submitted transaction hash=0xd25314b188adf3f45b05ebfec54f01b3c9e0afaa5
3fab76645a8f93c36629637 from=0x9cdaeba50e1e7df1690b80385dd5FEA8d7c8be22 nonce=3 recipient=0x62b80603cc312da015d6519
4ceb6f5dbd73E0696 value=1,000,000,000,000,000,000
"0xd25314b188adf3f45b05ebfec54f01b3c9e0afaa53fab76645a8f93c36629637"
> miner.start()
INFO [01-30|14:20:23.845] Updated mining threads threads=2
INFO [01-30|14:20:23.845] Transaction pool price threshold updated price=1,000,000,000
null
> INFO [01-30|14:20:23.846] Commit new mining work number=3 sealhash=a20981..5260ad uncles=0 txs=
0 gas=0 fees=0 elapsed="254.907us"
INFO [01-30|14:20:23.847] Commit new mining work number=3 sealhash=d3c146..b4bb55 uncles=0 txs=4
gas=84000 fees=0.000483 elapsed=1.022ms
INFO [01-30|14:20:41.081] Successfully sealed new block number=3 sealhash=d3c146..b4bb55 hash=f132af..56
d006 elapsed=17.234s
INFO [01-30|14:20:41.085] mined potential block number=3 hash=f132af..56d006
INFO [01-30|14:20:41.091] Commit new mining work number=4 sealhash=1e18e5..96bcb8 uncles=0 txs=0
gas=0 fees=0 elapsed=4.203ms
```

26. Now start the miner

```
miner.start(1)
```

27. Stop the miner after it has mined an block and check the balance

```
miner.stop()
eth.getBalance("0x62b80603cc312da015d65194ceb6f5dbd73e0696")
```

Updated Balance from 0 to 1114000000000004000


```
Activities Terminal Jan 30 14:25 ganesh@ganesh-VirtualBox: ~/singlenode

INFO [01-30|14:25:19.445] Successfully sealed new block
INFO [01-30|14:25:19.445] block reached canonical chain
INFO [01-30|14:25:19.445] mined potential block
INFO [01-30|14:25:19.445] Commit new mining work
apssd="129.652µs"
INFO [01-30|14:25:21.132] Successfully sealed new block
INFO [01-30|14:25:21.132] block reached canonical chain
INFO [01-30|14:25:21.132] mined potential block
INFO [01-30|14:25:21.137] Commit new mining work
apssd="5.083ms"
INFO [01-30|14:25:22.194] Successfully sealed new block
INFO [01-30|14:25:22.195] block reached canonical chain
INFO [01-30|14:25:22.195] mined potential block
INFO [01-30|14:25:22.195] Commit new mining work
apssd="173.56µs"
INFO [01-30|14:25:22.719] Successfully sealed new block
INFO [01-30|14:25:22.719] block reached canonical chain
INFO [01-30|14:25:22.719] mined potential block
INFO [01-30|14:25:22.720] Commit new mining work
apssd="142.44µs"
INFO [01-30|14:25:26.207] Successfully sealed new block
INFO [01-30|14:25:26.207] block reached canonical chain
INFO [01-30|14:25:26.207] mined potential block
INFO [01-30|14:25:26.208] Commit new mining work
apssd="537.583µs"
> INFO [01-30|14:25:28.774] Successfully sealed new block
INFO [01-30|14:25:28.774] block reached canonical chain
INFO [01-30|14:25:28.774] mined potential block
INFO [01-30|14:25:28.774] Commit new mining work
apssd="308.358µs"
> eth.getBalance("0x62b80603cc312da015d65194ceb6f5dbd73e0696")
11140000000000000000
> INFO [01-30|14:25:30.947] Successfully sealed new block
INFO [01-30|14:25:30.947] block reached canonical chain
INFO [01-30|14:25:30.963] mined potential block
INFO [01-30|14:25:30.963] Commit new mining work
apssd="16.508ms"

number=112 sealhash=cb9320..9bf544 hash=b3f5de..43aa57 elapsed=336.738ms
number=105 hash=b86cc8..61a896
number=112 hash=b3f5de..43aa57
number=113 sealhash=0f59e9..7c0a01 uncles=0 txs=0 gas=0 fees=0 el
number=113 sealhash=0f59e9..7c0a01 hash=6731d0..5380a1 elapsed=1.686s
number=106 hash=fcc4af..e113d2
number=113 hash=6731d0..5380a1
number=114 sealhash=1f31e9..167b2e uncles=0 txs=0 gas=0 fees=0 el
number=114 sealhash=1f31e9..167b2e hash=d63674..af7461 elapsed=1.056s
number=107 hash=223a8c..806711
number=114 hash=d63674..af7461
number=115 sealhash=9bcc53..955b87 uncles=0 txs=0 gas=0 fees=0 el
number=115 sealhash=9bcc53..955b87 hash=0fccc5..a0af42 elapsed=524.362ms
number=108 hash=472aa1..f09b40
number=115 hash=0fccc5..a0af42
number=116 sealhash=5f519a..879fde uncles=0 txs=0 gas=0 fees=0 el
number=116 sealhash=5f519a..879fde hash=a2e9c4..3f86aa elapsed=3.486s
number=109 hash=34f6a4..a99857
number=116 hash=a2e9c4..3f86aa
number=117 sealhash=7d5839..81603a uncles=0 txs=0 gas=0 fees=0 el
number=117 sealhash=7d5839..81603a hash=dbd13f..0c799f elapsed=2.566s
number=110 hash=3e8e01..a4104c
number=117 hash=dbd13f..0c799f
number=118 sealhash=1f857f..0fb962 uncles=0 txs=0 gas=0 fees=0 el
number=118 sealhash=1f857f..0fb962 hash=8add4e..3679a5 elapsed=2.172s
number=111 hash=9cdd83..61e76d
number=118 hash=8add4e..3679a5
number=119 sealhash=539e50..9d1bef uncles=0 txs=0 gas=0 fees=0 el
```